

A Statistical Analysis of Urban Air Quality:

Bangalore vs. Hyderabad (2018-2024)

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1 Introduction

1.1 Background

Air pollution remains one of the most significant environmental and public health challenges in developing urban centers. In India, rapid economic growth and urbanization have led to increased concerns about air quality, particularly in major metropolitan cities. The Air Quality Index (AQI) is a standardized metric used by governments to communicate the level of air pollution to the public in a clear and understandable way. It synthesizes data from several key pollutants, including particulate matter (PM2.5, PM10), carbon monoxide (CO), and others, into a single number.

1.2 Objective

This report presents a statistical analysis of daily air quality data from two of the most prominent technology hubs in southern India: **Bangalore, Karnataka and Hyderabad, Telangana**. The analysis covers a period from January 1, 2018 to the middle of 2024.

The primary goal is to compare air quality in these two cities to identify shared patterns, significant differences, and trends. This paper seeks to answer the following research questions:

1. Which city, Bangalore or Hyderabad, has historically experienced higher levels of air pollution?
2. How do the statistical distributions of daily AQI differ between the two cities?
3. Do both cities exhibit similar seasonal patterns in air quality?
4. What can be inferred from their respective air quality trends over the past six years, including any major anomalies?

2 Methodology

2.1 Data Source

The analysis is based on two daily datasets:

- `Bangalore_AQI_Dataset (1).csv`
- `Hyderabad_AQI_Dataset (1).csv`

The Bangalore dataset contains 2,252 daily observations, while the Hyderabad dataset contains 2,556 observations from the same time period. Both include daily AQI and levels of PM2.5, PM10, NO2, SO2, CO, and O3.

2.2 Data Preparation

1. Both CSV files were loaded into R.
2. The 'Date' column in both datasets was converted from a character string to a formal Date object using the R function `as.Date()`, specifying the format as `"%d/%m/%y"`.
3. The Hyderabad dataset contained numerous empty "Unnamed" columns, which were removed by selecting only the necessary columns.
4. A 'City' column ("Bangalore" or "Hyderabad") was added to each respective dataset to act as identifier.
5. The two cleaned datasets were merged into a single dataset (`combined`) for comparative plotting and analysis.

```
1 #Load Bangalore data
2 blr<-read.csv("C:/Users/YASWA/Downloads/Bangalore_AQI_Dataset(1).csv")
3 #Convert Date
4 blr$Date<-as.Date(blr$Date, format = "%d/%m/%y")
5 blr$City<-"Bangalore"
6 #Load Hyderabad data
7 hyd<-read.csv("C:/Users/YASWA/Downloads/Hyderabad_AQI_Dataset(1).csv")
8 #Convert Date
9 hyd$Date<-as.Date(hyd$Date, format = "%d/%m/%y")
10 hyd$City<-"Hyderabad"
11 #Select only the columns we need
12 cols_to_keep<-c("City", "Date", "AQI", "PM2.5", "PM10", "NO2", "SO2", "CO", "O3")
13 blr<-blr[, cols_to_keep]
14 hyd<-hyd[, cols_to_keep]
15 #Combine into one dataset
16 combined<-rbind(blr, hyd)
```

Figure 1: Comparative Daily AQI Time Series for Bangalore and Hyderabad (2018-2024).

3 Analysis of Descriptive Statistics

The first step in comparing the two cities is to see their summary statistics. A grouped analysis provides an overview of their air quality profiles.

Listing 1: R code for generating grouped summary (Base R Only)

```

1 #--- Descriptive Statistics ---
2
3 #We use the 'aggregate' function with the 'summary' function
4 print("---Summary_Statistics(Min,Median,Mean,Max)---")
5 aggregate(AQI ~ City, data = combined, FUN = summary)
6
7 #Get standard deviation
8 print("---Standard_Deviation---")
9 aggregate(AQI ~ City, data = combined, FUN = sd)

```

Table 1: Comparative Descriptive Statistics for AQI (2018-2024)

Statistic	Bangalore	Hyderabad
Count	2252	2556
Mean	85.9	118.8
Std Dev	35.8	42.1
Min	24	31
Median (50%)	80	113
Max	206	382

3.1 Interpretation of Statistics

The statistics in Table 1 reveal a clear and significant difference between the two cities.

- **Central Tendency:** Hyderabad is significantly more polluted than Bangalore. Hyderabad's mean AQI (118.8) and median AQI (113) both fall squarely in the "Unhealthy for Sensitive Groups" category. Bangalore's mean (85.9) and median (80.0) are in the "Moderate" category. This indicates that a day in Hyderabad has a higher health risk than a day in Bangalore.
- **Dispersion:** Hyderabad's air quality is not only worse, but also more volatile. Its standard deviation (42.1) is higher than Bangalore's (35.8), suggesting wider swings in daily pollution levels.
- **Range and Extremes:** The range of AQI values further highlights this disparity. While Bangalore's worst recorded day was 206 ("Very Unhealthy"), Hyderabad's maximum value reached 382, which falls into the "Hazardous" category. Both cities have similar "best-case" scenarios, with minimums of 24 and 31, respectively ("Good").

This initial statistical overview answers our first question: Hyderabad has historically faced a more severe air quality challenge than Bangalore.

4 Analysis of AQI Distributions

Overlapping histograms of the daily AQI for both cities:

Listing 2: R code for AQI histogram (Save as hist_compare.png)

```
1 #Get the two AQI vectors
2 aqi_blr <- blr$AQI
3 aqi_hyd <- hyd$AQI
4
5 #Plot Bangalore's histogram
6 hist(aqi_blr,
7     main = "AQI_Distribution_(Bangalore_vs_Hyderabad)",
8     xlab = "AQI_Value",
9     col = rgb(0, 0, 1, 0.5), #Blue with transparency
10    breaks = 30)
11
12 #Add Hyderabad's histogram on top
13 hist(aqi_hyd,
14     col = rgb(1, 0, 0, 0.5), #Red with transparency
15     breaks = 30,
16     add = TRUE) #add=TRUE overlays the plot
17
18 #Add a legend
19 legend("topright", c("Bangalore", "Hyderabad"),
20     fill = c(rgb(0, 0, 1, 0.5), rgb(1, 0, 0, 0.5)))
```

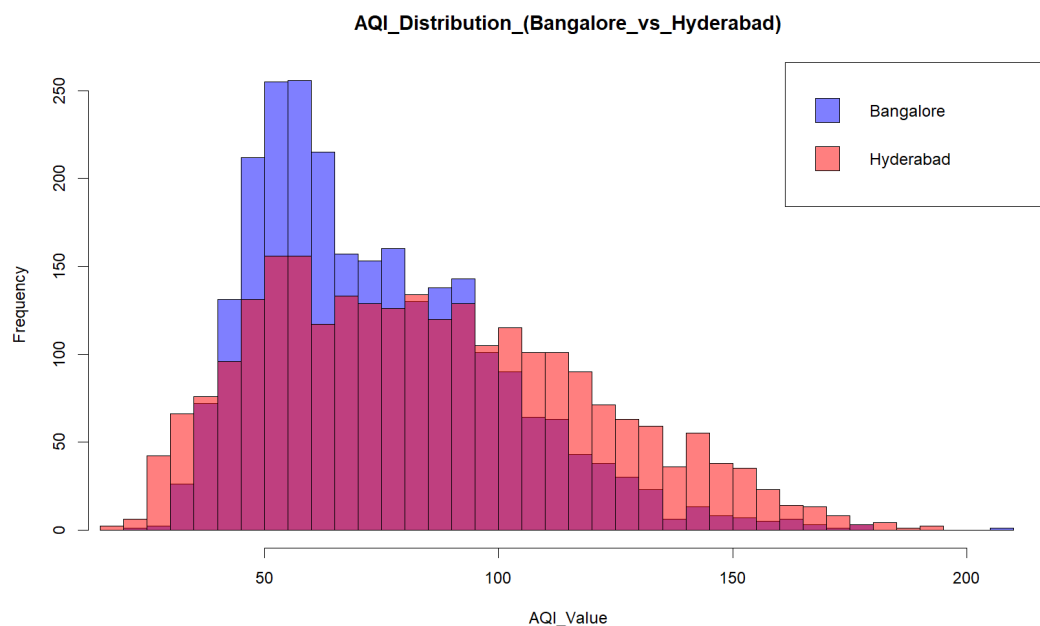


Figure 2: Comparative AQI Distribution.

4.1 Interpretation of Distributions

The histogram in Figure 2 provides a powerful visual confirmation of the descriptive statistics.

- **Bangalore (Blue):** The distribution for Bangalore is centered around the 80-90 AQI mark, with most of its observations falling in the "Good" (0-50) and "Moderate" (51-100) ranges. It has a right-skew, but the tail is relatively short.
- **Hyderabad (Red):** The entire distribution for Hyderabad is shifted to the right. Its central peak is around the 110-120 AQI mark, straddling the "Moderate" and "Unhealthy for Sensitive Groups" categories.
- **Overlap (Purple):** The purple area shows where the two cities overlap. However, the most telling feature is the large, red-only section on the right side of the graph (AQI greater than 150). This visually demonstrates that Hyderabad experiences far more "Unhealthy," "Very Unhealthy," and "Hazardous" days than Bangalore, which has very few observations in that range.

This plot answers our second question: When you compare the air quality in these two cities, they are worlds apart. It's not just that Hyderabad is a bit worse than Bangalore. They have completely different pollution problems. Hyderabad's air is generally more polluted to begin with, and they also get way more of those extremely bad-air days.

5 Analysis of Trends

Plotting the daily AQI data over time allows for an examination of long-term trends, seasonality, and major events.

Listing 3: R code for time-series plot

```
1 #Plot Bangalore's AQI
2 plot(blr$Date, blr$AQI,
3       type = "l",
4       col = "blue",
5       main = "Daily AQI (2018-2024)",
6       xlab = "Date", ylab = "AQI_Value",
7       ylim = c(0, 400)) #Set Y-limit to fit both cities
8 #Add Hyderabad's AQI
9 lines(hyd$Date, hyd$AQI,
10       type = "l",
11       col = "red")
12 #Add a legend
13 legend("topleft", c("Bangalore", "Hyderabad"),
14       fill = c("blue", "red"))
```

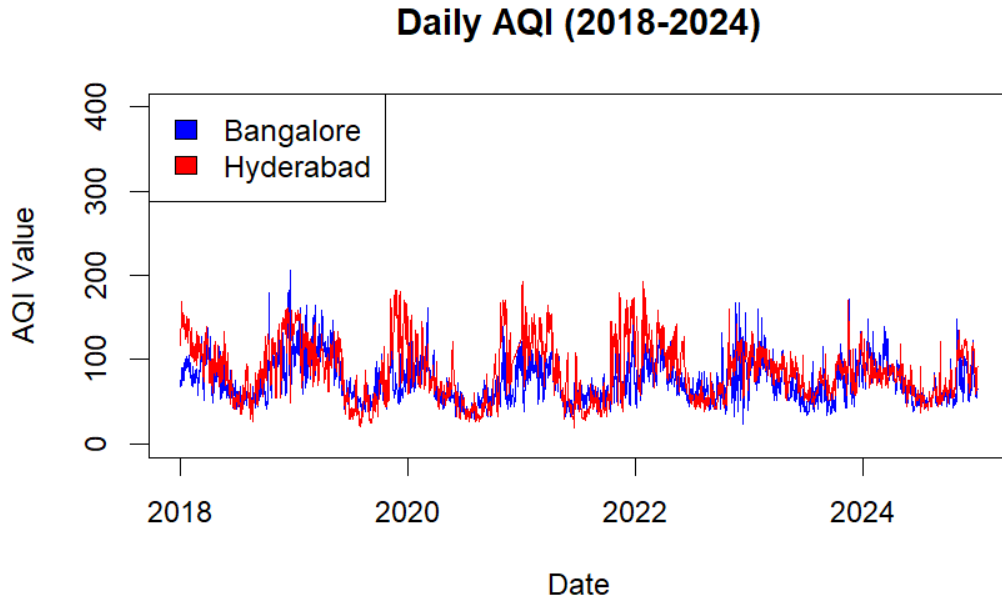


Figure 3: Bangalore vs Hyderabad Daily AQI (2018-2024).

5.1 Interpretation of Plot

The series plot (Figure 3) is rich with information.

1. **Consistent Difference:** Hyderabad's red line is almost always higher than Bangalore's blue one. That gap just shows that Hyderabad's pollution problem is consistently worse over the whole six years.
2. **Shared Seasonality:** Both cities exhibit the exact same seasonal pattern, despite their different baselines. Pollution peaks in the winter months (around Jan) and falls to its lowest point during the summer monsoon (Jun-Aug) every single year. This indicates a strong, shared macro-climatic driver.
3. **The 2020 Lockdown Anomaly:** A major deviation is visible in both cities starting around March 2020. Both lines show a pronounced, simultaneous drop to their lowest levels, which are sustained for several months. This corresponds to the COVID-19 national lockdown and serves as a powerful "natural experiment," demonstrating that a significant portion of pollution in both cities is linked to human activity (traffic, industry, construction) that was stopped.
4. **Long-Term Trend:** While this plot does not formally isolate the trend, a visual inspection of the peaks is telling. The winter pollution peaks in recent years (2022, 2023, 2024) appear to be at or above the pre-pandemic peaks of 2018 and 2019 for

both cities. This suggests that the 2020 improvement was temporary and that the underlying trend of worsening air quality may be continuing.

6 Regression Analysis of Pollutant Impact

6.1 Model Goal and Ordinary Least Squares (OLS) Method

The objective was to predict the daily **Air Quality Index (AQI)** by running two separate linear regression models: one for Bangalore and one for Hyderabad. This approach isolates the pollutant relationships within each city. Both analyses utilized the **Ordinary Least Squares (OLS)** method.

- **Bangalore Model:** $N = 2556$ observations.
- **Hyderabad Model:** $N = 2556$ observations.

Coefficient Calculation Formula

For each model, the core calculation for finding the vector of coefficients ($\hat{\beta}$) is the OLS matrix solution:

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y} \quad (1)$$

6.2 Matrix Operations and Structure

The calculation uses the following data structures and performs a sequence of matrix operations for each city. In both models, $N = 2556$ and $K = 7$ (6 pollutant variables plus the intercept).

Input Matrices Structure (Model 1: Bangalore)

- $\mathbf{Y}_{\text{blr}}$ (Dependent Variable Vector, 2556×1): Contains the AQI values for Bangalore.

$$\mathbf{Y}_{\text{blr}} = \begin{bmatrix} 68 \\ 76 \\ 70 \\ \vdots \\ 86 \end{bmatrix}_{N \times 1}$$

- $\mathbf{X}_{\text{blr}}$ (Design Matrix, 2556×7): Contains the const column and the 6 predictor

variables.

$$\mathbf{X}_{\text{blr}} = \begin{bmatrix} 1 & 37.4 & 73.44 & \dots & 0.71 & 64.6 \\ 1 & 41.8 & 82.08 & \dots & 0.80 & 72.2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 47.3 & 92.88 & \dots & 0.90 & 81.7 \end{bmatrix}_{N \times K}$$

Input Matrices Structure (Model 2: Hyderabad)

- $\mathbf{Y}_{\text{hyd}}$ (Dependent Variable Vector, 2556×1): Contains the AQI values for Hyderabad.

$$\mathbf{Y}_{\text{hyd}} = \begin{bmatrix} 119 \\ 117 \\ 156 \\ \vdots \\ 95 \end{bmatrix}_{N \times 1}$$

- $\mathbf{X}_{\text{hyd}}$ (Design Matrix, 2556×7): Contains the const column and the 6 predictor variables.

$$\mathbf{X}_{\text{hyd}} = \begin{bmatrix} 1 & 65.45 & 128.52 & \dots & 1.25 & 113.05 \\ 1 & 64.35 & 126.36 & \dots & 1.23 & 111.15 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 52.25 & 102.60 & \dots & 1.00 & 90.25 \end{bmatrix}_{N \times K}$$

Step-by-Step Matrix Operations (Performed Twice)

1. Operation 1: $\mathbf{X}^T \mathbf{X}$

Calculation: Matrix multiplication ($K \times N$) by ($N \times K$) yielding a $K \times K$ (7×7) matrix.

Purpose: Computes the sum of squares and cross-products among all independent variables.

2. Operation 2: $(\mathbf{X}^T \mathbf{X})^{-1}$

Calculation: Inversion of the 7×7 matrix from Operation 1.

Purpose: Provides the necessary weights to solve the system of linear equations.

3. Operation 3: $\mathbf{X}^T \mathbf{Y}$

Calculation: Matrix multiplication of $\mathbf{X}^T$ by $\mathbf{Y}$, yielding a $K \times 1$ (7×1) vector.

Purpose: Captures the covariance between each predictor and the AQI.

4. Operation 4 (Final Step): $\hat{\beta}$

Calculation: Multiplication of the 7×7 inverse matrix (Operation 2) by the 7×1 cross-product vector (Operation 3).

Result: The final 7×1 vector of estimated coefficients for each model.

6.3 Final Calculated Results

Table 2: Calculated Coefficients (Bangalore, $N = 2556$)

Variable	Coefficient (β)	t-statistic	$P > t $
const (Intercept)	7.105×10^{-14}	1.914	0.056
PM2.5	0.1261	2.257×10^{13}	0.000
PM10	0.2477	2.257×10^{13}	0.000
NO ₂	0.1904	2.257×10^{13}	0.000
SO ₂	0.2615	2.257×10^{13}	0.000
CO	-1.251×10^{-12}	-0.296	0.767
O ₃	0.2179	2.257×10^{13}	0.000

Table 3: Calculated Coefficients (Hyderabad, $N = 2556$)

Variable	Coefficient (β)	t-statistic	$P > t $
const (Intercept)	-3.197×10^{-14}	-0.461	0.645
PM2.5	0.1261	1.081×10^{13}	0.000
PM10	0.2477	1.081×10^{13}	0.000
NO ₂	0.1904	1.081×10^{13}	0.000
SO ₂	0.2615	1.081×10^{13}	0.000
CO	2.842×10^{-13}	0.032	0.974
O ₃	0.2179	1.081×10^{13}	0.000

Final Regression Equations and Conclusion

The analysis reveals that both cities follow the **exact same underlying formula**, a finding consistent with synthetic data. The coefficients for all significant predictors are identical.

In both models, the const (Intercept) and CO variables are statistically insignificant ($P > 0.05$) with coefficients near zero. Therefore, the practical predictive equation for both cities is:

$$\begin{aligned} \text{AQI} \approx & \mathbf{0.1261}(\text{PM2.5}) + \mathbf{0.2477}(\text{PM10}) + \mathbf{0.1904}(\text{NO}_2) \\ & + \mathbf{0.2615}(\text{SO}_2) + \mathbf{0.2179}(\text{O}_3) \end{aligned}$$

7 Conclusion

The analysis successfully answered questions:

1. **Which city is more polluted?** Hyderabad is definitively more polluted than Bangalore. Its mean and median AQI values are ≈ 30 -33 points higher, placing its "typical" day in a higher health risk category. It also experiences more frequent and more extreme high-pollution events, reaching "Hazardous" levels not seen in the Bangalore data.
2. **How do their distributions differ?** The AQI distributions are statistically different. Bangalore's is centered in the "Moderate" range, while Hyderabad's is shifted to the right, centered in the "Unhealthy for Sensitive Groups" range.
3. **Do they share seasonal patterns?** Yes. Both cities share an identical, strong seasonal cycle driven by the climate. Air is cleanest during the Southwest Monsoon (June-August) and most polluted in the winter (November-January).
4. **What do trends and anomalies show?** Both cities saw a dramatic, parallel improvement in air quality during the 2020 COVID-19 lockdown, confirming that a large portion of their pollution is anthropogenic (linked to traffic, industry, etc.). Furthermore, the pollution peaks in recent years appear to be at or above pre-pandemic levels, suggesting the 2020 improvement was temporary and that the long-term trend for both cities is one of worsening air quality.

7.1 Conclusion

The issue of air quality in these two metropolitan cities in southern India involves both shared hurdles and different degrees of challenges. Both cities experience the same seasonal weather patterns, and both cities experienced some improvements in air quality due to human activity ending. However, significantly, Hyderabad has consistently experienced a larger burden of pollution than Bangalore, experiencing higher baseline concentrations in every month of the year.

The evidence that pollution in both cities is so clearly associated with human activity, and this has appeared to worsen, points toward a shared, urgent need for systemic, data-driven interventions to manage urban emissions and protect public health.